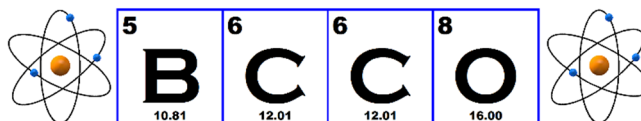


2025 JUNIOR CANADIAN CHEMISTRY OLYMPIAD

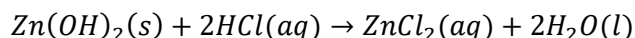
British Columbia Chemistry Olympiad



The Periodic Table/Data Sheet may be used for all parts of the contest, loose paper, pen, pencils and non-programmable calculators are allowed.

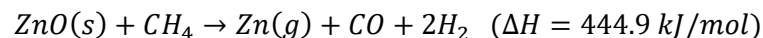
Students are expected to attempt all questions for a complete examination in 1.5 hours, it is recognized that backgrounds will vary and students will not be eliminated from further competitions because they have missed parts of the paper.

- 1) What is the type of the following reaction?



- a) Synthesis
 - b) Single replacement
 - c) Decomposition
 - d) Neutralization**
 - e) Combustion
- 2) The density of zinc chloride at 20 °C is 2.907 g/cm³. What is the volume of 1 mole of zinc chloride at 20 °C?
- a) 59.1 cm³ b) 34.7 cm³ **c) 46.9 cm³** d) 136 cm³ e) 396 cm³
- 3) Which of the following equations is not properly balanced?
- a) $2\text{CH}_3\text{OH} + 3\text{O}_2 \rightarrow 2\text{CO}_2 + 4\text{H}_2\text{O}$
 - b) $3\text{NaOH} + \text{H}_3\text{PO}_4 \rightarrow \text{Na}_3\text{PO}_4 + 2\text{H}_2\text{O}$**
 - c) $\text{C}_{12}\text{H}_{22}\text{O}_{11} + \text{H}_2\text{O} \rightarrow 2\text{C}_6\text{H}_{12}\text{O}_6$
 - d) $4\text{C}_3\text{H}_5\text{N}_3\text{O}_9 \rightarrow 12\text{CO}_2 + 10\text{H}_2\text{O} + 6\text{N}_2 + \text{O}_2$
 - e) $4\text{NH}_3 + 3\text{O}_2 \rightarrow 6\text{H}_2\text{O} + 2\text{N}_2$
- 4) Potassium permanganate (KMnO₄) is dark purple, almost black. It is an oxidising agent with disinfectant, deodorising, and astringent properties. How many grams of potassium permanganate is required to prepare a 0.1 M solution in a 200 mL volumetric flask?
- a) 0.316 b) 15.8 c) 2.20 d) 11.0 **e) 3.16**

- 5) Solar energy is a plentiful, clean, and sustainable energy source, which can contribute to protect the environment. Hydrogen is also an important sustainable energy option, but clean hydrogen production can be problematic. 50% of the world's hydrogen production is done by methane reforming. Methane reforming is a chemical reaction consisting in extracting dihydrogen from methane, the reaction is endothermic and therefore requires heat. A promising method of methane reforming uses a solar-driven approach in combination with zinc oxide:



If a total of 87 million tons of dihydrogen are produced in one year, how many million of tons of carbon monoxide are produced? Assume that methane reforming is the sole source of carbon monoxide.

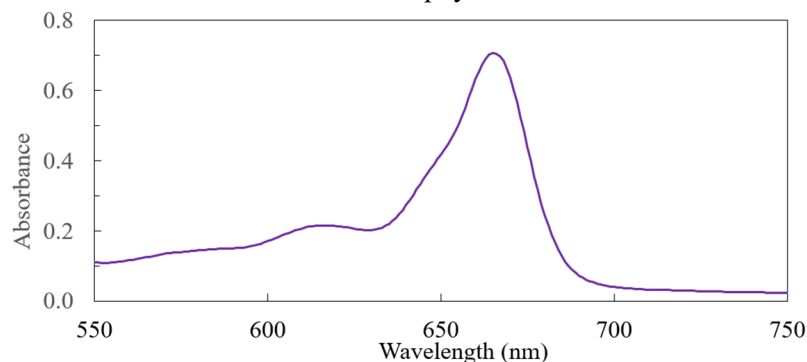
- a) 302 b) 43.5 c) 174 d) 87 **e) 604**
- 6) When a colorless solution A is added to a sodium chloride solution, a white precipitate is formed. The precipitate is soluble in ammonia. The electrolysis of A gives a shiny metal that is a very good electricity conductor. Identify the metal in solution A.
- a) Na b) Fe c) Al **d) Ag** e) Cu

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- 7) In UV/Vis spectroscopy, the Beer–Lambert law equation relates the absorbance, the molar absorptivity of the molecule, the path length of the light, and the concentration of the solution as follows:

$$A = \epsilon lc$$

Where A is absorbance, ϵ is the molar extinction coefficient (a property of the molecule), and l is the path length of the light through the sample, and c the concentration of the solution. A cell with an optical path of 1 cm was filled with a solution of chlorophyll a in acetone. The molar extinction coefficient of chlorophyll a is $7.9 \times 10^4 \text{ M}^{-1} \text{ cm}^{-1}$ at 665 nm. The spectrum recorded is shown below. What is the concentration of chlorophyll a ?



- a) $1.33 \times 10^{-8} \text{ mol/L}$
b) $8.86 \times 10^{-6} \text{ mol/L}$
 c) $1.27 \times 10^{-5} \text{ mol/L}$
 d) $8.42 \times 10^{-3} \text{ mol/L}$
 e) $1.05 \times 10^{-3} \text{ mol/L}$
- 8) The Athabasca basin, located in northern Saskatchewan and Alberta, provides the highest grade of uranium ore in the world. The region supplies about 20% of the world's uranium. The half-life of uranium-238 is about 4.5 billion years. What is the rate constant for the uranium-238 decay?
- a) $2.9 \times 10^{-16} \text{ s}^{-1}$
 b) **$4.2 \times 10^{-13} \text{ day}^{-1}$**
 c) 0.15 yr^{-1}
 d) $4.2 \times 10^{-4} \text{ day}^{-1}$
 e) $2.2 \times 10^{-10} \text{ yr}^{-1}$

- 9) In a 1 L glass container, solid calcium carbonate reacts with hydrogen chloride gas as follows:



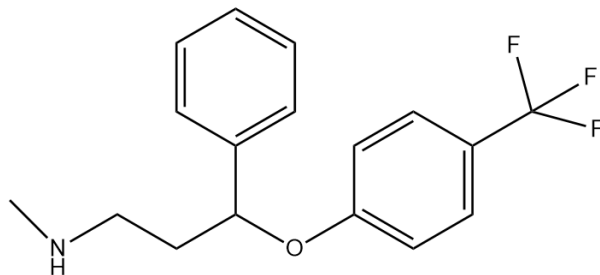
Initially the glass container is filled with 10 atm of HCl gas and 10 g of calcium carbonate only. Select the statement that is true if this reaction is carried at 25°C.

- a) HCl is the limiting reagent
 b) 22.2 g of CaCl_2 will be produced.
 c) The final pressure will be higher than the initial pressure
d) CaCO_3 is the limiting reagent
 e) 0.198 moles of CO_2 will be produced
- 10) In assuming ideal gas behavior for the reaction in question 9, what would be the final pressure in the container at 25°C?
- a) 7.58 atm** b) 2.42 atm c) 0 atm d) 5.15 atm e) 12.4 atm
- 11) Effective nuclear charge (Z_{eff}) is a useful tool for understanding periodicity. Z_{eff} is the net positive charge experienced by an electron in a polyelectronic atom. The term "effective" is used because the positive charge of the nuclei cannot reach a given electron completely due to the shielding effect of other electrons. Z_{eff} is a theoretically calculated value. A good estimate of Z_{eff} can be obtained using Slater's rule. Slater's rule states that $Z_{\text{eff}} = Z - S$, where Z is the atomic number of the atom and S a screening constant. The screening value is 0.85 for a core electron and 0.35 for a valence electron. The screening constant (S) is obtained by the summation of all the screening values of other electrons, excluding the electron for which the effective nuclear charge is to be determined. Use Slater's rule to estimate the Z_{eff} for a valence electron in a hydride anion (H^-).
- a) 1.00 b) 0.15 c) 1.65 d) -1.00 **e) 0.65**

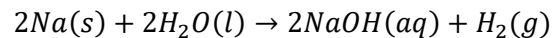
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- 12) The structure of an antidepressant that is on the World Health Organisation's List of Essential Medicines is shown below. This Medicine is prescribed to more than 24 million people in the USA. Which of the following molecular formula is the correct one for the antidepressant shown below?

- a) $C_{17}H_{17}F_3NO$
 b) $C_{17}H_{18}F_3NO$
 c) $C_{16}H_{15}F_3NO$
 d) $C_{18}H_{18}F_3NO$
 e) $C_{18}H_{17}F_3NO$



- 13) A piece of sodium is added to water in a beaker and fire can be observed. The reaction that takes place is:



Predict the signs of ΔH , w and ΔU . (Answers given in this order).

- a) $- / - / -$
 b) $+ / - / +$
 c) $- / + / +$
 d) $+ / + / +$
 e) $- / - / +$

- 14) A diamond is the hardest natural substance on earth, but if it is placed in an oven and the temperature is raised to about 763°C , it will simply vanish, without even ash remaining. Only a little carbon dioxide will have been released. If the heats of combustion of diamond and graphite are -395.4 kJ/mol and -393.5 kJ/mol respectively, what is the heat of formation of diamonds?
- a) -1900 J/mol b) 788.9 kJ/mol c) -788.9 kJ/mol
 d) -0.002 kJ/mol e) 1900 J/mol

- 15) How much energy is needed to heat 30mL of water from 5° to 30°C ? ($C_{p,s} = 4.18\text{ Jg}^{-1}\text{K}^{-1}$)?

- a) 1881 J b) 174.2 J c) 3135 J d) 104.5 J e) 38000 J

- 16) Following the VSEPR theory, which of the following molecules has a molecular geometry that is trigonal bipyramidal?

- a) ClF_3 b) PF_2Cl_3 c) NH_3 d) XeOF_4 e) BrF_5

- 17) Which of the following atoms has the highest first ionization energy?

- a) F b) He c) Cl d) Ar e) Br

- 18) Which of the following species has the smallest radius?

- a) H^- b) He c) Li^+ d) Be^{2+} e) B^{3+}

- 19) Which of the following double replacement reactions will form a yellow precipitate?

- a) $\text{AgNO}_3(\text{aq}) + \text{NaCl}(\text{aq})$
 b) $\text{AgNO}_3(\text{aq}) + \text{KI}(\text{aq})$
 c) $\text{AgNO}_3(\text{aq}) + \text{NaCH}_3\text{COO}(\text{aq})$
 d) $\text{AgNO}_3(\text{aq}) + \text{K}_2\text{CrO}_4(\text{aq})$
 e) $\text{AgNO}_3(\text{aq}) + \text{K}_2\text{SO}_4(\text{aq})$

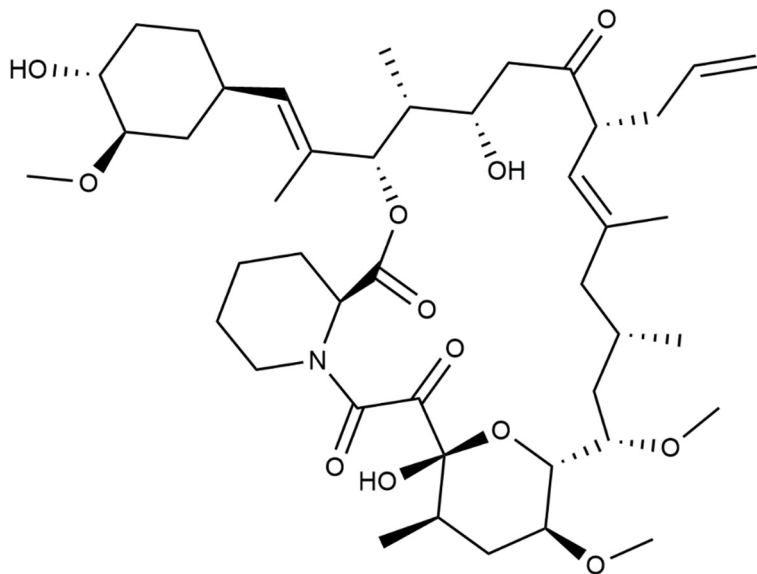
- 20) Dimethyl ether and ethanol are organic isomers, they have the same

- I. Molar mass.
 II. Normal boiling point.
 III. Surface tension.
 IV. Density at 20°C and 1 atm.
 V. Density at 100°C and 1 atm.

- a) I and II b) I and III c) I and IV
 d) I and V e) I, IV and V

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- 21) Tacrolimus is a drug used after organ transplant to lower the risk of organ rejection. The structure is shown below. Which of the following functional groups the molecule contains?



- a) Alkene, alcohol, amine, and ketone
 b) Alkyne, alcohol, amine, and ether
 c) Alkene, amide, ester, and ketone
 d) Aldehyde, amide, ester, and ketone
 e) Alkene, amine, ether, and ester
- 22) Each of the following 1 liter of aqueous solution is prepared by dissolving 0.1 moles of each of the compounds in a sufficient amount of water. Which solution will have the highest percentage of free ions for all ions both at pH = 0 and pH = 14?
- a) NaCl and KNO₃
 b) NH₄Cl and NaCH₃COO
 c) CuSO₄ and NaCl
 d) Na₂S and NaCl
 e) Na₂CO₃ and Ca(NO₃)₂

- 23) Which of the following is a valid set of quantum numbers for the unpaired electron in ground state copper atom?

- a) 3, 1, 0, $-\frac{1}{2}$ b) 3, 2, 2, $\frac{1}{2}$ c) 3, 3, -1, $-\frac{1}{2}$
 d) 4, 0, 0, $\frac{1}{2}$ e) 4, 1, 0, $\frac{1}{2}$

- 24) Thermal decomposition of a sample of 1.25 grams of HgO containing unreactive impurities produces 56.35mL of oxygen gas collected at 22°C under standard atmospheric pressure. What is the percent purity of the sample? (Hint, the products are mercury and oxygen gas).

- a) 40.3% b) 49.2% c) 89.2%
 d) 80.6% e) 98.4%

- 25) Metalloid elements are key to the microelectronics industry. Microchips used in microelectronics are made by silicon, germanium and other metalloid elements. N-type semiconductors are made by

- a) Doping Silicon with Germanium
 b) Doping Silicon with Arsenic
 c) Doping Silicon with Boron
 d) Doping Arsenic with Silicon
 e) Doping Germanium with Silicon

End of examination questions.

	5 B 10.81	6 C 12.01	6 C 12.01	8 O 16.00	
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